

Project name: Shop AI - The Digital Waiter for Restaurants

Team name & members (id, name): MFU 888

- 6931503001 Aung Nyi Nyi Thant
- 6931503031 Chawakron Singkaew
- 6931503042 Tharathon Plengsri
- 6931503073 Sasiwimon Chatkaew
- 6931503103 Natthanon Wongsai

PROBLEM

- **The problem, in 2–3 sentences:** Small-to-midsize independent restaurants often rely on staff to take orders and answer customer questions. During peak hours or when understaffed, this causes slow service, order mistakes, and delays in answering common questions about ingredients, opening hours, and menu recommendations.
- **Who exactly has it (not “everyone”):** Front-of-house staff—servers, hosts, and cashiers—especially in restaurants with limited staff, busy peak periods, or new/part-time employees who may not know the menu well. The menu and can’t confidently answer questions about ingredients or allergens.
- **Why it matters / what it costs them today:** Order mistakes waste food, kitchen time, and money, while slow responses frustrate customers and can lead to lower tips and negative reviews. Staff also spend valuable time answering repetitive questions instead of focusing on customers who need personal assistance.

GATE 1 — REACHABLE USERS

- **May – Front-of-House Employee at a Made-to-Order Restaurant —**
 - *Interviewed on September 22, 2026, at a restaurant in Chiang Rai; approximately 15 minutes*
 - May works at the front of the restaurant alongside two other staff members. During the peak period from 12:00 to 13:00, approximately 15–20 customers may arrive at the same time. When it gets very busy, she has to take orders, serve food, clear tables, and answer customers’ questions simultaneously.
 - She said: “Sometimes several tables call us at the same time, so we have to rush to do everything. As a result, sometimes orders get missed.”
 - **Key Insight:** Customers frequently become dissatisfied when staff cannot immediately answer menu or ingredient questions during high-volume periods.
- **Chef Kung – Chef/Owner of a Noodle Restaurant —**
 - *Interviewed on September 22, 2026, at a noodle restaurant in Chiang Rai; approximately 14 minutes*
 - Chef Kung runs the restaurant with does not have any serving staff. As a result, he has to prepare food, take orders, and answer customers’ questions by himself.
 - Special requests, such as “**no fresh pork or beef blood**,” are among the orders he is most likely to get wrong when the restaurant is busy. Preparing a new bowl of noodles takes approximately 5–7 minutes and adds around 20–30 baht in additional cost per bowl.
 - He explained: “When there are a lot of customers, I have to remember many things at once. Sometimes a customer tells us not to add fresh pork blood, but after we finish making the dish, we realize that we forgot.”
 - **Key insight:** He is concerned that using a tablet-based ordering system or an automated ordering system could make the service feel less personal, especially for regular customers who are accustomed to ordering and talking directly with the owner.
- **P’Nut – Owner of a Fried Food Shop —**
 - *Interviewed on September 22, 2026, at a fried food restaurant in Chiang Rai; approximately 20 minutes*
 - P’Nut has three employees, which is not enough to handle the volume of customers during approximately 5:30–7:00 PM. The employees have to repeatedly answer the same general questions, especially during peak hours when many customers are waiting for their food. This sometimes leads to incorrect orders—for example, a customer may request a non-spicy dish but receive a spicy one. The staff then have to prepare a new dish, resulting in additional time being spent by both the restaurant and the customer.
 - He explained: “Sometimes, problems occur during the order-taking process because we are in too much of a hurry. Customers may say several things at the same time, which makes it difficult for us to remember everything.”
 - **Key insight:** Employees have to handle both in-store customer orders and phone orders at the same time, which increases the risk of miscommunication and order errors.

GATE 2 — MVP SHAPE

- **3–5 planned functional requirements**
 - FR-1: Multilingual Menu Browsing — The system shall provide a QR menu in Thai, Burmese, and English.
 - FR-2: AI Waiter Chat — answer customer questions using the restaurant’s data.
 - FR-3: Order & Staff Call — allow customers to select dishes, send orders, and call staff.
 - FR-4: Live Staff & Kitchen Screens — show orders and staff calls in real time.
 - FR-5: Menu & Allergen Management — allow owners to manage menus, prices, and allergen information.
 - FR-6: Menu Import — allow owners to import menu information from a photo.
 - FR-7: Restaurant Settings — allow owners to manage opening hours, FAQs, specials, and AI settings.
 - FR-8: Table QR Codes — generate a unique QR code for each table.
 - FR-9: Staff Accounts — allow owners to create staff accounts with roles and PINs.
 - FR-10: Restaurant Insights — show owners customer questions and unanswered topics.
- **Which FR carries a real rule or decision (not plain CRUD)**
 - FR-2 (AI Waiter Chat) — needs logic to decide how to answer using restaurant data, and handle ambiguous questions — not just reading/writing data
 - FR-10 (Restaurant Insights) — needs to *determine* which topics are “unanswered,” requiring aggregation and analysis, not a direct query
 - FR-6 (Menu Import) — needs logic to convert a photo into structured menu data (OCR + parsing decisions)
- **What data has to persist between sessions?**
 - Menu, prices, allergen info (FR-5)
 - Orders and order status (FR-3, FR-4)
 - Restaurant info: opening hours, FAQs, specials, AI settings (FR-7)
 - Per-table QR codes (FR-8)
 - Staff accounts, roles, PINs (FR-9)
 - Log of customer questions/unanswered topics for insights (FR-10)

GATE 3 — ACCESS

- **Any data, licence or system you depend on — and how you know you can get it**

Everything runs on one laptop with no paid services.

- AI model: the open-weight gemma4:12b model runs locally through Ollama, which is free to download and needs no API key.
- Menu data: comes from the restaurant owner, either typed in or read from a photo of the printed menu.
- Database: SQLite, which is built into Node.js.
- No outside systems: no POS or payment integration, no cloud AI API.
- No personal data: diners don’t log in, and no personal details are stored

PROCESS & TEAM Chosen SDLC (with a 1-line reason): Agile (iterative sprints): the AI’s safety answers and its Burmese replies needed repeated test-and-fix rounds (a 30-question test set and 4 reviews by a native Burmese speaker), and the screens are refined from user feedback every sprint.